

## Numerical Methods

BCA — Semester 3 — Answer Sheet

Subject Code:

**Note:** This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

### Group B

**1. Write an algorithm and a C-Program to obtain roots of non-linear equation using Newton Raphson Method.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**2. Solve the following ordinary differential equation using shooting method.  $(y'' + xy' - xy = 2x)$ , with boundary conditions  $(y(0)=1)$ ; and  $(y(2)=10)$**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**3. Compare and contrast between Jacobi iterative methods and Gauss Seidal method? Solve the following equation using Gauss Seidal method.  $(x + 2y + 3z = 5)$ ,  $(2x + 8y + 22z = 6)$  and  $(3x + 22y + 82z = -10)$**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**4. Use secant method to estimate the root of the equation  $(x^2 - 5x + 6 = 0)$ , with initial estimate  $(x_1 = 4)$  and  $(x_2 = 2)$  (EPS=0.05).**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**5. Solve the double integration using Simpson's 1/3 rule.  $\int_2^{2.6} \int_4^{4.4} \frac{dx dy}{xy}$**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**6. What are the sources of errors? Discuss various types of errors encounters in numerical computation.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**7. Fit a second order polynomial to the data in the table below:**

|   |    |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|----|
| X | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| Y | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

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8. Solve the following differential equation  $\frac{dy}{dx} = 3x + \frac{y}{2}$  with  $y(0) = 1$  for  $x = 0.2$  ( $h = 0.1$ ) using Euler's Method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Why Numerical Integration is required? Compute the integral:  $I = \int_{-1}^1 e^x dx$  using composite trapezoidal rule for  $n = 4$ .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Evaluate  $\frac{dy}{dx}$  at  $x = 5$  using Newton's forward interpolation formula using the following table.

| x | y        |
|---|----------|
| 1 | 1.2012   |
| 2 | 1.80119  |
| 3 | 2.60472  |
| 4 | 3.801302 |
| 5 | 5.280    |

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Find the Eigen values and Eigen vectors of the Matrix:  $A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Solve the Poisson's equation  $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 2x^2y^2$  over the square domain  $(0 \leq x \leq 3)$  and  $(0 \leq y \leq 3)$  with  $f=0$  on the boundary and  $h = 1$ .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

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These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.